Build and test your reasoning

Page 1 | A: 0-15 min. B: 15-40 min. Keep original attempts and add corrections separately.

Start at week03-study.html. Follow the English video and its pauses. Plan about 180 minutes including calculations, Labs 03/04, reflection and a 10-minute break. Timing is guidance, not a requirement to wait. Write here or in a notebook using the same labels. Background reading is optional.

A. Retrieve the VLE ideas

A1. Define x_1, y_1, z_1 and γ_1. Distinguish liquid composition from liquid activity a_1.

A2. Use $\gamma_1=y_1P/(x_1P_1^{sat})$, with $y_1=0.704857144627$, $P=158.707198211$ kPa, $x_1=0.50$ and $P_1^{sat}=150$ kPa. Compare with exp(0.399813). State the vapor and pressure-correction assumptions.

A3. A model fits the data closely. State what this establishes and one question it does not settle.

B. Connect the equation to the experimental path

B1. At fixed T and P, differentiate $g=x \ln \gamma_1+(1-x)\ln \gamma_2$ using the product rule. Identify the term removed by $x\, d \ln \gamma_1+(1-x)d \ln \gamma_2=0$. Show why $dg/dx=\ln(\gamma_1/\gamma_2)$. Here g means $g^E/(RT)$.

B2. Complete the table before checking guide page 1. Name a required correction or assumption, not simply a test name.

Path / available information	Local diagnostic: correction or limitation
Fixed T and P, composition varies	
Isothermal VLE, equilibrium P varies	
Real isobaric VLE, no independent h^E	

B3. Explain the difference between “not applicable,” “not assessed” and “outside this criterion.” May the first two be counted as passed?

Output: one derivation, a path table and your corrected A answers. Next: calculations on page 2.

Calculate global and local evidence

Page 2 | C: 40-65 min. Use a calculator or spreadsheet; show substituted values and units or dimensionless labels.

C1-C3. Area calculation

x_1	0	0.25	0.50	0.75	1
$L=\ln(\gamma_1/\gamma_2)$	2	1	0	-1	-2

C1. Compute all four trapezoids, positive area, signed negative area and signed sum S. **C2.** Add 0.20 to every L and recalculate S. **C3.** Explain why a small integral cannot certify every observation. These synthetic endpoint limits are supplied analytically. S is not normalized D.

C4. Follow the worked local calculation

x	$\ln \gamma_1$	$\ln \gamma_2$	$g=x \ln \gamma_1+(1-x)\ln \gamma_2$
0.40	0.72	0.32	
0.50	0.50	0.50	
0.60	0.32	0.72	

At x=0.50, calculate $L=\ln \gamma_1-\ln \gamma_2$ and $r\approx[g(0.60)-g(0.40)]/0.20-L(0.50)$. Then change **only $\ln \gamma_1$ at x=0.60 from 0.32 to 0.36** and repeat. Follow the worked explanation; retain the clean and changed values.

C5. Your independent local calculation

x	$\ln \gamma_1$	$\ln \gamma_2$	g
0.30	0.735	0.135	
0.40	0.540	0.240	
0.50	0.375	0.375	

Calculate r at x=0.40 using the same **g-first** procedure. Repeat after changing **only $\ln \gamma_2$ at x=0.50 from 0.375 to 0.425**. Show the two slopes and residuals before opening the check.

C6. Both local tables are synthetic fixed-T,P examples with uniform spacing. Why can spacing, rounding and noise change a finite-difference result? Why is this one-point r not the Lab 03 score 100 mean |r|?

Predict, run, compare and explain

Page 3 | D/E: 65-95 min. Guide pages 2-3. After this block, take the 95-105 min break.

Open Lab 03: Consistency tests

D1. Follow the real baseline demonstration, then reproduce **Synthetic · isothermal, 350 K / Consistent reference / 41 points**. Inspect area and local charts; record and save the reference. Do not fit yet.

D2-D3: predict BEFORE changing the control or viewing a check. For canceling errors and a localized defect, predict which diagnostic may notice the discrepancy and why. An uncertain prediction is acceptable; preserve it.

Case	Your prediction and reason
D2. Canceling, amplitude 0.25	
D3. Localized, amplitude 0.25	

Now run **D2: Canceling errors · zero net area** and **D3: Localized defect near $x_1 = 0.35$**, both at amplitude 0.25 / 41 points. **D4:** keep the localized defect and change only sampling to **161 points**. Record actual statuses, including unexpected ones.

Run	Area D (%) / status	Local score / status	Residual shape / peak location
D1. Clean / 41			
D2. Canceling / 41			
D3. Localized / 41			
D4. Localized / 161			

D5. Return to Consistent reference / 41 points. Run **Run modeling test →** once; record Δp in % and Δy in percentage points. Explain why a close fit is not a general consistency certificate and why Herington is not applicable to this path.

E1. Save or sketch a local-residual plot. Annotate the axes, case, amplitude, sample count and important peak or oscillation. Compare D3/D4: distinguish a numerical sampling change from new experimental evidence.

E2. Write 3-4 sentences: claim, numerical/chart evidence, and one assumption or limitation. Revisit both predictions and explain any correction. Use a separate page if needed.

Keep: comparison table, annotated plot, explanation and separate exports for clean, canceling, localized-41 and localized-161. Save the clean fit before changing inputs. Export is local, not submission.

Predict liquid stability before calculating

Page 4 | F/G: 105-155 min. Complete F and G1 here before recording the Lab results on page 5.

F. Candidate equilibrium, stable equilibrium and amounts

F1. A solver finds equal chemical potentials. What additional global condition is needed for stable equilibrium? State the fixed conditions and material-balance constraint.

F2. For a supplied supported tie line with $x^{\alpha}=0.20$ and $x^{\beta}=0.80$, calculate $f\beta=(z_1-x^{\alpha})/(x^{\beta}-x^{\alpha})$ and $f\alpha=1-f\beta$ for $z_1=0.50$ and 0.35. Verify both component-1 balances. These hand endpoints are not Lab 04 outputs.

F3. Explain why positive curvature at the feed, or a line that touches twice, does not alone prove global stability.

G1. Three predictions, one fixed model

Use **Margules · 1 parameter, A=3**. Predict the homogeneous-feed state and equilibrium phase count for each z_1 below **before opening a named preset**. State your reason or uncertainty. The named presets reveal the feed classification.

z_1	Predicted homogeneous feed	Predicted equilibrium / reason
0.02		
0.12		
0.50		

A homogeneous-feed label concerns a hypothetical single liquid. The final equilibrium may have a different phase count. The model represents fixed hypothetical conditions, with no T/P controls.

Next: open Lab 04 and guide page 4. Keep A fixed while changing z_1. Test the predictions, rather than replacing them with copied labels.

Test stability and close the balances

Page 5 | Continue F/G to minute 155. Guide page 4 gives exact Lab 04 controls and export steps.

Open Lab 04: Liquid stability and LLE

G2. Run A=3 at z_1=0.02, 0.12 and 0.50. For each run record the homogeneous-feed label separately from the final phase count, curvature sign and Gibbs-energy decrease / RT. For a one-liquid result, report its composition and mark coexistence fractions not applicable.

z_1	Feed label / curvature sign	Final phase count / energy decrease	Actual x^{α}, x^{β} and $f\beta$ (if two liquids)
0.02			
0.12			
0.50			

G3. Using the actual Lab endpoints, independently calculate fβ, fα and $f\alpha x^{\alpha}+f\beta x^{\beta}$ for z_1=0.12 and 0.50. Show substituted numbers. Why is applying these two endpoints to z_1=0.02 infeasible?

G4. Energy and global support. Define $m(x)=\Delta g_{mix}/(RT)=x \ln x+(1-x)\ln(1-x)+3x(1-x)$. At z_1=0.12, calculate $m(z_1)$, $m_{split}=f\alpha m(x^{\alpha})+f\beta m(x^{\beta})$, and $m(z_1)-m_{split}$. Use your calculated fractions and Lab endpoints. This m includes ideal mixing; C used excess energy g only.

Annotate a sketch or screenshot of the supporting tangent: mark x^{α}, x^{β} and z_1=0.12. Explain how positive local curvature can coexist with a lower-energy finite split. Check distinct endpoints, chemical-potential agreement and global support in **Verify the equilibrium**. Compare predictions with observations.

Keep: the three-case table, two balances, energy calculation, annotated tangent and explanations. Save each run with **Export study JSON**, naming z002, z012 and z050.

Optional, only if time remains in this block: at z_1=0.50 compare A=1.8, 2 and 3. This changes the model, unlike changing the feed. No A sweep, NRTL comparison, LLE fit or VLLE task is required.

Transfer, correct and reflect

Page 6 | H: 155-170 min. I: 170-180 min. Attempt all unseen cases before viewing their solutions.

H. Use the ideas on new cases

H1. Synthetic fixed-T,P data give g(0.20)=0.32, g(0.30)=0.42, g(0.40)=0.48 and L(0.30)=0.75, where $g=g^E/(RT)$. Calculate the central slope and r at 0.30. Does one r determine a full-range consistency verdict?

H2. A real isobaric dataset gives a raw local score of 8.4, but independent h^E is unavailable. Should the local method be recorded as failed, passed or not assessed? Explain.

H3. Candidate coexistence endpoints are 0.15 and 0.75; z_1=0.39. Calculate both liquid fractions and the balance. What further evidence would establish that this candidate is stable?

H4. A hypothetical full-range isothermal assessment reports D=1.2%, local score=8.4, Δp=0.3% and Δy=0.2 percentage points. Use the Lab teaching criteria to classify each available test. Explain why a single overall “passed” label is misleading.

After attempting H1-H4: open worked solutions

Compare, correct in a different color and retain your first answers. Keep the residual, average score and fitting errors distinct.

I. Exit ticket, feedback and saved evidence

170-180 min: use the existing Session 3 form during the pause on slide 43. Answer its LLE **concept** question using your three-state investigation. In the **evidence** answer include your Lab 03 area/local comparison, settings, numerical results and one limitation.

Complete feedback, including **actual study minutes and any pacing difficulty** in the existing feedback text. Submit once; wait for **Response received**. On slide 44, verify the receipt and save the record, corrections and Lab files/screenshots. Do not resubmit. The form is anonymous, not an individual attendance record. Lab exports and this worksheet do not submit themselves.

Session 3 exit ticket + feedback

Scan the QR or use the clickable link below. The form is anonymous. It is not an attendance-identification form.

Open the Session 3 form